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RESEARCH ARTICLE

Automated Vehicle Regulation Needs to Speak to Code, not to Humans: Keeping Safety and Ethics in the Public Domain

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Abstract

In anticipation of the market introduction of highly and fully automated vehicles, regulations for their behavior in public road traffic are emerging in various countries and regions. Yet, as we show using the example of EU and German regulations, these rules are both incomplete and exceptionally vague. In this paper we introduce three traffic scenarios highlighting conflicting ethical, legal, and utility-related claims, and perform a legal analysis with regards to the expected behavior of AVs in these scenarios. We show that the existing regulatory framework disregards the realities of algorithmic decision-making in automated vehicles, such as the incomplete and imprecise perception of their environment and the probabilistic nature of their predictions. Importantly, the current regulations are written in abstract language addressing human interpreters rather than the precise logical-numerical computer code. We argue that the required interpretation and translation of the abstract legal language into the logical-numerical domain is so ambiguous that the regulations as they stand fail to guide or limit automated vehicle behavior in any meaningful way. This comes with significant ethical implications, as the interpretation and translation is unavoidable and, if not provided by regulatory bodies, will have to be performed by manufacturers. We argue that ethical decisions with significant impact on public safety must not be delegated to private companies, and thus, regulatory frameworks need significant improvements.

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1 Introduction

Trajectory decisions in automated vehicles (AVs) are guided by implicit and explicit assumptions about safety, fairness, and responsibility in road traffic, reflecting both ethical and legal values. As one and the same trajectory planning algorithm can be expected to run on an entire vehicle fleet at the same time, the logic and values embedded within this algorithm will have direct implications for the safety and risk exposure of virtually all road users, mandating comprehensive regulation and oversight.

Explicit debates on the ethics of AVs have only been given serious attention for about 10 years (Goodall, 2014), and legal debates for slightly longer (for an overview, see Nyholm, 2018). An important stream of the current ethical debate discusses moral preferences of laypeople and their role in the regulation of AVs. The prime examples of this are The Moral Machine Experiment (Awad et al., 2018) and related studies (Sütfeld et al., 2017; Faulhaber et al., 2019; Sütfeld et al., 2019; Bigman & Gray, 2020; Paulo & Möck, 2024). These are inspired by the so-called 'Trolley Problem', a thought experiment developed in the 60s and 70s dealing with moral dilemmas and especially the doctrine of double effect, where an apparently good course of action has a morally negative side effect (Foot, 1967; Thomson, 1976). In fact, such thought experiments are much older as legal scholars have been discussing hypothetical scenarios much earlier. Josef Kohler discussed in 1915 a case with the title Autolenker-Fall, the "case of the car driver" (for an overview, see Kirchmair, 2023c). What is rather new is that these thought experiments received a very real case of application. AVs are on the rise, and they provoke important ethical and legal issues (Kirchmair, 2023a). A specific ethical issue is the question as to how to regulate car crashes with AVs (Kirchmair, 2023b).

However, as we will show with the example of the EU type approval regulation for motor vehicles (inlcuding AVs)[1] and the German Road Traffic Regulation (in German Straßenverkehrs-Ordnung, StVO)[2], current laws fail to address many ethical dimensions of AV trajectory planning, and are often exceptionally vague in those they do address. This stands in sharp contrast with the computer code in the vehicle that

[1] Regulation (EU) 2018/858 of the European Parliament and of the Council of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles, OJ L 151, 14.6.2018, p. 1-218; amended by Regulation (EU) 2019/2144 of the European Parliament and of the Council of 27 November 2019 on type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards their general safety and the protection of vehicle occupants and vulnerable road users, OJ L 325, 16.12.2019, p. 1-40; see further Commission Implementing Regulation (EU) 2022/1426 of 5 August 2022 laying down rules for the application of Regulation (EU) 2019/2144 of the European Parliament and of the Council as regards uniform procedures and technical specifications for the type-approval of the automated driving system (ADS) of fully automated vehicles (Text with EEA relevance), OJ L 221, 26.8.2022, p. 1-64.

[2] Straßenverkehrs-Ordnung (StVO), of 6 March 2013 (BGBl. [Federal Law Gazette] I p. 367), in German available at https://www.gesetze-im-internet.de/stvo_2013/BJNR036710013.html.

deterministically controls its behavior, executing logical and numerical operations with absolute precision. Bridging this gap requires an interpretation and/or translation of the abstract legal language into logical and numerical operations. The vagueness in the language of the law and the absence of numerical quantification, then, give rise to a high degree of interpretative freedom. In other words, current laws provide very little concrete guidance regarding the decision-making logic implemented in the vehicle, leaving both risk-utility trade-offs and the distribution of risk between various road users largely in the hands of the private manufacturers. As Krügel and Uhl (2024) point out, the notion of numerical risk-utility trade-offs is not merely an academic hypothetical: "In a series of patents[13–15], Google Inc. and later Waymo LLC, for instance, set out how AVs can evaluate different actions based on a risk-cost framework. This framework adopts the expected-utility criterion, which compares available actions based on the respective sum of risk penalties, each determined as the product of risk magnitude and probability".

Our main contributions to the current debate on regulations of AV behavior is to highlight the central tension between the current state of the law and the realities of trajectory-related decision-making in AVs. We evaluate the current laws on AVs in the EU and Germany and show that they fail to effectively regulate the behavior of AVs. We argue that a logically-numerically anchored legal framework, explicitly regulating ethical cost/benefit calculations is missing and that this is a severe problem. We show this by introducing three traffic scenarios illustrating conflicting ethical, legal and utility-related claims where each scenario highlights at least one potential omission or critical imprecision in the current regulatory framework.

Some clarifications and definitions are needed before jumping into the scenarios. These set the stage and delimit the scope of the paper.

First, we use the term "trajectory" to describe the result of a series of steering and acceleration/deceleration commands executed along a path, encompassing both the exact points along this path and the associated speed of the vehicle at each point. "Trajectory planning" then encompasses all trajectory-related decisions and entails the consideration of all ethically and legally relevant factors connected to these decisions.

Second, for "risk", we follow the convention of the risk management-related literature, where it is typically defined as probability times severity or, more generally, cost of undesirable events (Goodall, 2016). We take risk to include both personal injury as well as property damage, unless specified otherwise. The term "utility" is used in this paper to encompass all non-risk-related consequences of a chosen trajectory, such as passenger comfort and enjoyment, fuel or energy efficiency, and impact on estimated travel time and traffic flow.

Third, we consider trajectory decisions to have an ethical dimension when they incur any trade-off between risk of personal injury, risk of property damages, expected utility, or a potential violation of traffic rules, including trade-offs within individual categories (e.g., a trade-off of the risk of personal injuries for multiple involved parties). This understanding of ethics encompasses virtually all kinds of trajectory-related decisions, including mundane ones like adapting the vehicle's target speed based on the given traffic situation, or deciding on its lateral position in the lane (Himmelreich, 2018; Krügel & Uhl, 2024; Cecchini et al., 2023). We thus view almost any action pertaining to the vehicle's speed, location or direction of travel to entail an expression

of some degree of risk-tolerance, and we further regard any ethical implications and their significance to be a feature of the situation in which a decision is made, not of the decision-making process (Klenk, 2021; Verbeek, 2006). The failure to consider ethically relevant aspects of a situation, thus, does not diminish the ethical relevance of said situation.

Fourth, in order to keep the arguments focused we assume a modular system architecture for automated driving, such as the sense-plan-act control methodology, which is the standard system architecture for automated driving systems to date (International Organization for Standardization (ISO), 2022; Shalev-Shwartz et al., 2017; Gruyer et al., 2017; Sormoli et al., 2024). Here, the planning stage is functionally separate from the perception and execution stages. We also assume the planning stage to be manually implemented (e.g., Poszler et al., 2024; Geisslinger et al., 2023), as opposed to, for instance, machine learning (ML) approaches (e.g., Cheng et al., 2024). However, this distinction isn't central to our thesis, and we address the topic of ML approaches for planning and decision making, including end-to-end systems, separately in the discussion section.

Fifth, our way of viewing ethics in this paper cuts across the major ethical theories of deontology, consequentialism, and virtue ethics. These ethical theories are various accounts of what is, or explains, the morally good and bad, or right and wrong (Singer, 2023). For the purposes of this paper we are agnostic regarding this discussion. The perspective in this paper is similar to the Ethical Valence Theory proposed in Evans et al. (2020), which "paints AV decision-making as a type of claim mitigation: different road users hold different moral claims on the vehicle's behavior, and the vehicle must mitigate these claims as it makes decisions about its environment." Ethical claims, in our view, can take into account any number of factors, such as the foreseen outcome of an action, distribution of risk, behavioral rules, and anything else that is relevant to what constitutes ethically and legally right behavior, including traffic rules and property values. We thus do not take a stance on what factors are or are not relevant in the decision-making process (with few exceptions), and do not advocate any particular programming of AVs, nor do we criticize those who do (see for such proposals, e.g., Bonnefon et al., 2020; Luetge, 2017; as well as numerous academic works, as for instance, Himmelreich, 2018; Paulo, 2023; Rodríguez-Alcázar et al., 2021; Paulo & Kirchmair, 2025; Henschke & Arora, 2024). The central point in this paper is instead that current regulation of AV behavior is flawed, regardless of what is considered ethically correct behavior.

2 Scenario Analysis

In the following section we present three road traffic scenarios and evaluate current regulations in light of the ethical challenges the scenarios raise. We chose the regulations within the European Union (EU) in combination with the German law as the example case because, to the best of our knowledge, they are the most comprehensive regulations of AV behavior to date.

The EU type approval regulation and its implementing regulation (see fn. 1) feature uniform procedures and technical specifications for the type-approval of the automated

driving system (ADS) of fully automated vehicles, and will be the basis for our scenario analysis.

In the EU, the competence to enact laws for the technical vehicle regulation is within the responsibility of the EU, while driving behavior-related regulations are the responsibility of the member states. This traditional division is challenged by AVs, as an AV's technical properties give rise to its driving behavior – automotive technology and driving behavior are therefore no longer trivially separable. This may lead to a diffusion of responsibilities between the respective institutions, which in turn may hinder the realization of an appropriate regulatory framework for AVs. In some instances, technical regulation provided by the EU may require defined technical *abilities*, e.g., to perform emergency maneuvers (Annex II, Section 2. Commission Implementing Regulation (EU) 2022/1426), while the member states' road traffic acts determine the conditions under which these capabilities must be utilized as part of the vehicle's behavior. In other instances, however, the EU's technical regulation defines the expected behavior more directly (e.g. Annex II, Section 1.1.3. Commission Implementing Regulation (EU) 2022/1426) and explicitly addresses ethical elements of the decision-making process that precedes a trajectory decision (e.g. Annex II, Sections 2.1.1.1 and 2.1.1.2 Commission Implementing Regulation (EU) 2022/1426). Similar issues with the division of responsibilities might be found in non-EU jurisdictions as well.

The following three scenarios highlight specific conflicts of ethical, legal, and utility-related claims on trajectory decisions. They are simplified for illustrative purposes, set aside any problems that might occur at the perception stage (by assuming the system correctly identifies all relevant elements of the situation) and, where applicable, quantify the estimated likelihood of potential outcomes in percentage values. In the real world, an AV might more likely use a metric such as the expected velocity differential at time of collision, possibly coupled with a function that associates this differential with an estimated severity of injury for each of the involved road users. The number of trajectory options would most likely be higher, or trajectories might even be evaluated on a continuous spectrum. However, these simplifications do not affect the central conflicts portrayed in the following scenarios.

- ***Scenario One: The Stranded Vehicle.*** *An AV is driving in city traffic and encounters a vehicle that has broken down and is blocking the AV's lane, which is separated from the oncoming lane with a solid line. There is no immediate oncoming traffic. The AV has two options: (1) Move across the solid line to pass the vehicle. (2) Wait behind the vehicle until the situation is resolved.*

  Analysis: Firstly, the EU type approval regulation does not cite any specific requirements with regards to traffic rules other than the demand to "comply with traffic rules of the country of operation" (Annex II, Section 1.3 Commission Implementing Regulation (EU) 2022/1426). The German Road Traffic Regulation, in turn, considers a solid line a traffic sign (Annex 2, Section 1, 68, sign no. 295 German Road Traffic Regulation), the trespassing of which is a rule violation fined at 10 Euros if nobody is endangered, and more otherwise.
  In the given scenario, however, it could easily take half an hour or more until the lane-blocking situation is resolved, which is not only inconvenient for the

passengers of a single AV; with sufficiently many AVs on the road, an inability of AVs to cross the solid line could bring traffic in the entire area to a halt. Provided a perception and planning system that reliably determines a safe passing maneuver crossing over the solid line, rigid adherence to such a traffic rule would seem absurd: Following the law to the letter in this way produces severe disadvantages for all while simultaneously producing no increased safety or convienence. This seems undesirable no matter what ethical programming is considered preferable, as traffic rules aren't ends in themselves but are means to promote safety, efficiency, and fairness on the roads.

It is arguable that the basic rules in Section 1 of the German Road Traffic Regulation, and more specifically Section 1 para 1 and 2 apply. These state that participation in road traffic requires constant caution and mutual consideration, and anyone participating in road traffic must behave in such a way that no other person is harmed, endangered, hindered or inconvenienced more than is unavoidable under the circumstances. However, what is considered "unavoidable under the circumstances" is not defined, and it isn't clear how a potential conflict or contradiction between two rules should be resolved algorithmically. This rule can therefore not be translated into an algorithmic expression without further ado.

The scenario and its brief analysis illustrate that conflicts between traffic rules and utility (travel time, traffic flow) are common in urban traffic and important to address, and that rigid adherence to all specific traffic rules would lead to absurd behavior by AVs. It appears unavoidable that in edge-cases specific traffic rules need to be transgressed for reasons of utility, safety or both. While such commonsense decisions come naturally to human drivers, AVs will require a clear and precise definition of the circumstances under which this is permissible.

- ***Scenario Two: The High-Stakes Wager.*** *An AV is driving on a rural road when in a turn an oncoming car appears to cross over into the AV's lane. While the car is likely to course-correct, the AV predicts a 10% chance of the other car coming over into the AV's lane, potentially causing a head-on collision. The AV's options are: (1) Initiate an emergency breaking maneuver and stay in the lane, leading to a 10% chance of a head-on collision with severe injuries expected for both the AV's occupant and the driver of the oncoming car, as well as 50.000 EUR in material damages. (2) Initiate an emergency breaking maneuver and swerve into a ditch, leading to no expected injuries but expected material damages of 10.000 EUR.*

  Analysis: First, Annex II rule 2.1.1. Commission Implementing Regulation (EU) 2022/1426 applies in this scenario: "the AV shall be able to detect the risk of collision with other road users or a suddenly appearing obstacle (debris, lost load)", which in this scenario is the oncoming car appearing to lose control.

  Second, after detecting such a risk, the AV "shall be able to automatically perform appropriate emergency operation (braking, evasive steering) to avoid reasonably foreseeable collisions and minimise risks to the safety of the vehicle occupants and other road users". It is worth highlighting that the regulation only requires the AV to *be able to* perform emergency maneuvers, and that such emergency operations need to serve the purpose of avoiding collisions and minimizing risks. Clarifications regarding the circumstances under which emergency operations can or must be performed, are left to Member States' road traffic legislation.

Third, German law, namely the provision on collision avoidance systems in Section 1e para. 2 No. 2 of the German Act on Autonomous Driving[3], states that motor vehicles with an autonomous driving function must have an accident avoidance system that "(a) is designed to avoid and reduce harm, (b) in the event of unavoidable alternative harm to different legal interests, takes into account the importance of the legal interests, with the protection of human life having the highest priority;". As the laws are currently formulated, any consideration of probability is missing from both the EU and national law and it seems unclear how an AV should make decisions in the face of probabilistic or unknown consequences. In the scenario, the conditions that would trigger an evasive maneuver are directly linked to the valuation of human bodily integrity versus property damage, as both of them represent generally valid legal interests. The weighing up of these factors seems unavoidable, yet neither law defines any mechanism for this.

The term "prioritization" is also vague. The cited German Act on Autonomous Driving states that *the highest* – not *absolute* – priority shall be given to human *life* – not *bodily integrity* in general. Thus, it seems entirely unclear which level of "prioritization" (or *valuation*) is to be awarded at which level of expected injury. The two extreme interpretations – *absolute priority at any level of expected injury* (implying that a 1% risk of a scratch on the arm weighs larger than a million Euros in certain material damages) and *highest priority only in cases of certain death* – demarcate the range of possible interpretations of these sections. The large range of possible interpretation clearly requries greater clarification.

A comprehensive good-faith interpretation of the given laws is that "when a collision is foreseeable, the AV shall take evasive action with the aim of minimizing harm to human life, while balancing valid legal interests of all involved parties". However, this interpretation raises a number of follow-up questions: What does "foreseeable" mean, i.e., at what percentage of risk of a collision is a specific outcome deemed a "foreseen" one, which legal interests are valid under which circumstances, which other factors should or should not be considered in the situation, and most importantly, what constitutes correct or acceptable "balancing" or "weighting" in the eye of the law?

This example shows how seemingly simple rules formulated in abstract language quickly become complicated and require a lot of interpretation when an algorithmic implementation is required. In the absence of any regulatory interpretation or translation from the abstract-linguistic domain into the logical-numerical domain it would currently be up to the manufacturers to perform this interpretation, which may or may not align with public interest.

- ***Scenario Three: The Jaywalk Disaster.*** *A person unexpectedly runs into the street and in front of a small AV with two occupants. A truck is in the oncoming lane, both vehicles travel at 50 kph. There is free space on the sidewalk, with one otherwise uninvolved pedestrian some meters away. The AV is now facing three options: (1) Keep straight and perform emergency braking maneuver: After some speed*

[3] Gesetz zur Änderung des Straßenverkehrsgesetzes und des Pflichtversicherungsgesetzes - Gesetz zum autonomen Fahren, from 12 July 2021 (BGBl. [Federal Law Gazette] I p. 3108), in German available at https://www.gesetze-im-internet.de/stvg/BJNR004370909.html]Theenglishtranslationisbytheauthors.

reduction, there is a 50% chance of death of the pedestrian in front of the AV, no injuries expected for AV occupants. (2) Turn into oncoming traffic: 15% risk of death for each of the two AV occupants, no injuries expected for the truck driver. (3) Swerve onto sidewalk and perform emergency braking maneuver: 5% chance of death for the otherwise uninvolved pedestrian, no injuries for the AV occupants.

Analysis: In this scenario, the following rules apply:

1. Annex II Rule 2.1.1. Commission Implementing Regulation (EU) 2022/1426: "the AV shall be able to detect the risk of collision with other road users or a suddenly appearing obstacle (debris, lost load) and shall be able to automatically perform appropriate emergency operation (braking, evasive steering) to avoid reasonably foreseeable collisions and minimise risks to the safety of the vehicle occupants and other road users" – in this scenario, "other road users or a suddenly appearing obstacle" includes the person unexpectedly stepping into the street.
2. Annex II Rule 2.1.1.2. Commission Implementing Regulation (EU) 2022/1426: "The protection of other human life outside the fully automated vehicle shall not be subordinated to the protection of human life inside the fully automated vehicle." – And we assume that the inverse direction applies as well: Protection of life inside the vehicle shall also not be subordinated to protection of life outside of the vehicle. With this in mind, the rule could be read as "whether a person is inside or outside of the vehicle should have no bearing on the protection of their life."
3. Annex II Rule 2.1.2. Commission Implementing Regulation (EU) 2022/1426: "The vulnerability of road users involved should be taken into account by the avoidance/mitigation strategy."
4. German law, provision on collision avoidance systems in Section 1e para. 2 No. 2 of the German Act on Autonomous Driving: Motor vehicles with an autonomous driving function must have an accident avoidance system that "(a) is designed to avoid and reduce harm, (b) in the event of unavoidable alternative harm to different legal interests, takes into account the importance of the legal interests, with the protection of human life having the highest priority;".

Taken together, these rules demand that a collision avoidance system is present that can perform braking and evasive steering to avoid and/or mitigate collisions. This system shall minimize risks to people inside and outside of the vehicle (without favoritism towards one side or the other), take their respective vulnerability and legal interests into account, and prioritize the protection of human life above all else. What these rules do not provide is any kind of guidance on how to navigate a complex situation such as the one outlined in the scenario: Varying probabilities of potential outcomes (as discussed in scenario 2), varying levels of responsibility for causing the dangerous situation, any potential transgression of traffic rules (e.g., swerving into the oncoming lane), and potentially protected zones like, e.g., the sidewalk, remain unaddressed.

Regarding the consideration of responsibility or guilt as a factor in the decision, philosopher Harris pointedly asks whether we really would want a "death penalty

for jaywalking" (Harris, 2020). While the protection of human life clause seems to prohibit any draconic punishments, the law does not prohibit the consideration of responsibility in the vehicle's decision, nor does it set any concrete limits to the extent of its influence. Aside from the lack of guidance for the given scenario, there is another aspect worth mentioning. It could be argued that people inside of the AV have no influence on its perception, driving behavior, or ad-hoc decision-making, and should therefore bear no responsibility in any dangerous situation – even those caused by their AV – while people outside the AV may bear any amount of responsibility, as assigned by the AV itself. Partial blame in any amount on the side of a pedestrian may then be enough to tip the scales in favor of the AV passengers, effectively providing a backdoor for preferential treatment of the latter by manufacturers.

Regarding the permissibility of traffic rule transgressions, the scenario once again highlights the importance of context-specific considerations. Whether evasion onto the sidewalk (option 3) is a good solution in this scenario is debatable as it entails a small amount of risk to an otherwise uninvolved pedestrian, but at least in a variation of the scenario without any pedestrians on the sidewalk it becomes the clearly favorable option. The protection of human life clause seems to support this view: If the protection of human life has the highest priority, then the obeying of traffic rules – in this case the prohibition of driving on the sidewalk – cannot be of higher priority.

However, given the lack of specificity in the law, it again appears to be largely in the manufacturer's purview to decide upon the vehicle's behavioral principles and with that the outcome of scenarios like the one provided.

3 Discussion

The outlined scenarios illustrate some of the ambiguities that AVs will face in the real world, and therefore constitute a litmus test for any given legal framework governing trajectory-related decisions in AVs. Our analysis based on these scenarios highlights several severe flaws in the current state of the regulations in the EU, specifically Germany: (1) Rigid adherence to specific traffic rules can lead to unfavorable or even absurd outcomes but potential transgressions of these rules remain unaddressed. (2) An AV's perception and prediction of future events are inherently uncertain, yet the current regulations lack any guidance on how to deal with uncertainty in the decision-making process. And (3) many further aspects of decision-making remain unaddressed and tensions between existing rules remain unresolved. Our analysis shows that rules written for human drivers cannot be applied to AVs without further consideration, yet this exactly reflects the current state of the law. The failure to provide any AV-specific behavioral rules either on the level of the EU, or on the level of national traffic codes, is a glaring omission and results, in our opinion, in an untenable state of the law.

In the following discussion, we will address the shortcomings, omissions, and potential contradictions in the current law in more depth and elaborate on the need for a more precise and logically-numerically anchored framework for trajectory-decisions in AVs.

3.1 Traffic Rules and Proportionality

We begin with the topic of traffic rules and proportionality in the vehicles' decision-making. The rules for behavior in traffic to date are written for humans and demand the interpreter to apply their best judgment or common sense. Smaller rule violations are often necessary in order to keep traffic safe and flowing. Human drivers thus have the acknowledged ability to disobey the law when there appears to be no serious disadvantage to anyone, and would be implicitly expected to do so, e.g., in *The Stranded Vehicle* scenario. In order to enable similar behavior in AVs the specific circumstances must be explicitly specified for when the violation of a given traffic rule is permissible.

Rules can conflict with one another. For example, the rule "do not hinder or inconvenience others more than is unavoidable" and the rule "do not cross a solid line" can come into conflict. Most human drivers would likely resolve this conflict by crossing the solid line in a situation like the *The Stranded Vehicle* scenario above, if it is safe to do so. Presumably a driver that resolved the conflict in this way would not be punished by the law. In such a case, the severity of violating the specific rule (crossing a solid line) would be "weighed" against the broader safety and utility implications (preventing traffic jams and potential accidents). In practice, if crossing the solid line to pass the stranded car is the only reasonable and safe option, the general rule of not causing unnecessary hindrance would supersede the more special rule of not crossing the solid line. However, this way of resolving such norm conflicts is not a given for AVs because it is largely based on human intuition and post-factum analysis by experts (e.g., law enforcement or a judge). Evaluating such a situation relies heavily on the person's common sense, legal-ethical training and practice with similar cases. Further, there can be reasonable disagreement between experts: not all judges would agree on the verdict in all cases. Consequently there is no easy way for engineers to deduce which rules can be violated and when, from basic rules such as §1 of the German Road Traffic Regulation.

What this insight does for us is twofold: (1) It shows that a striving for reasonableness and proportionality is in place in the law as applied to human drivers, and (2) it highlights that this facet of the law and how it is applied in legal practice aren't necessarily captured by the rule book alone (in this case the German Road Traffic Regulation). Our conclusion from this is that AVs need their own rule book that specifies precisely when and how to perform a reasonableness test, or apply the principle of proportionality. It is precisely this that must be made explicit for AVs. In practice this means that traffic rules may need to be treated by the AV as soft constraints, as opposed to unconditional commandments. Some form of weighing or balancing, i.e., a proportionality test, has to be performed that decides when a specific rule is to be violated in the name of safety or utility (e.g., traffic flow).

Situations requiring such a proportionality test may arise suddenly and unexpectedly: Consider a version of *The Jaywalk Disaster* (scenario 3) without a second, uninvolved pedestrian on the sidewalk. Once the first pedestrian steps into the street, the vehicle has fractions of a second to make the decision to steer the car onto the empty sidewalk, thereby violating a traffic rule, in order to avoid a severe collision. Here, a proportionality test must be applied on the spot, it needs to be fast, and the test

may need re-evaluation at every step of the trajectory as the situation or the vehicle's perception of it changes. It might therefore make sense to think of the proportionality test as part of the regular trajectory planning routine, which equates to treating traffic rules as soft constraints: They are obeyed as long as this doesn't cause *unreasonable or disproportionate* harm, disruption of traffic, or inconvenience.

This argumentation then leads to the conclusion that the law needs to find a way to allow for traffic rule violations, but also a way to regulate under which circumstances these violations are permissible. It seems currently unclear whether an AV that is programmed to break traffic rules based on a proportionality test would get type approved or whether the manufacturers would expose themselves to any greater legal risk with such programming. For further reading on the topic of traffic rules and AVs see Reed et al. (2021).

3.2 The Realities of Algorithmic Decision-Making in AVs

The second topic needing more detailed discussion is the reality of algorithmic decision-making in AVs: The law should recognize that algorithms making trajectory-related decisions in AVs will inevitably make them based on an incomplete and imprecise perception of the situation and at best probabilistic predictions of future events, and that these algorithms will use logical-numerical computer code for this process (Sormoli et al., 2024; Chen et al., 2023).

A vehicle's perception systems, i.e., sensors and data processing pipeline, are inherently imprecise. Cameras, LiDAR, radar, and other sensors are used to detect all elements of the environment that are relevant to the driving behavior of the AV and other road users in the vicinity (for an overview see Chen et al., 2023). However, even in the best of conditions, margins of error exist around the exact position, direction of movement, and speed of any detected object. The detection of objects and classification into various object types is often flawed; pedestrians are only safely detected at fairly close distances, and the classification of less common modes of transportation (e.g., roller skates, skateboards, e-scooters), as well as animals and other roadside objects is often unstable. Moreover, a lot of the relevant information about the road and involved vehicles is only available as coarse estimates. This includes, e.g., surface type and grip levels, bumps, puddles, leaves, and ice patches on the road, as well as the state of tires and brakes, the weight and other characteristics of the involved vehicles with impact on brake distances, trajectory options and probabilities of collisions. Finally, occlusion and adverse weather conditions further decrease both precision and completeness of what the vehicle perceives of its environment and the current traffic situation.

In order to better judge the best possible actions and ideal trajectory in any given situation, intention recognition and the prediction of other road users' future trajectories may become part of an AV's trajectory planning stage (Huang et al., 2022). Some of this is easier; a heavy truck, for example, is quite limited in how fast it can change its speed or direction of travel, and under most circumstances has no reason to do so abruptly, while some of it is harder; a pedestrian realizing they walked in front of a moving vehicle might freeze, step back on the sidewalk, or start running

further across the street to get out of the way faster. Taking into account the previously described incomplete and imprecise perception of the directly observable current state, it becomes clear that predictions of future trajectories of other road users and objects are always going to be somewhat uncertain or probabilistic. As a result, the precise outcome of any given trajectory decision by the AV is going to be equally uncertain.

Given the imprecise and incomplete perception of the instantaneous situation, and the probabilistic nature of any projection into the future, any laws that simply prohibit or demand specific outcomes provide only vague guidance to the programming of decision-making algorithms. Take for example a rule that prohibits the killing of one person to save another: In reality, this trade-off will almost never present itself in a black and white fashion. Each trajectory option might instead contain some estimated amount of risk exposure to each of the people involved, with risk exposure being a function of probability of collision, expected speed differential at the moment of collision, vulnerability of the road user in question and further factors (Krügel & Uhl, 2024). The notion of killing or sacrificing one to save another also implies that one of the trajectory options is seen as the default, and the other as an alternative, which may or may not be plausible in any given situation. If, however, we accept the notion of a default option, the question then becomes whether or not a given risk exposure trade-off is justified. In other words, what amount of risk exposure decrease for one person (in the default trajectory) justifies a given risk exposure increase for another person (in the alternative trajectory). This question is potentially further complicated with additional factors such as responsibility for causing the dangerous situation, or being in a specific location that may grant special protections (e.g., the sidewalk). It is entirely unclear what amount of risk exposure increase to the person in the alternative trajectory by the AV's decision would be considered a sacrifice, i.e., illegal. The notion that a person cannot be killed in order to save another provides, therefore, very little guidance as to how the vehicle needs to be programmed to satisfy the law.

This example shows once more that ostensibly simple regulatory demands in abstract language are often far from straight forward to interpret and translate into logical-numerical rules. It is important to appreciate that (1) this interpretation and translation is unavoidable – the question is merely *who* performs it, and (2) that the vague formulations found in the current law allow for highly varied interpretations. In other words, the vehicle's actual behavioral policies are in large part for the manufacturers to decide, while the law, at least for the EU and Germany, does very little to limit what the vehicle can and can't do. While it is not the law's job to precisely define the way in which these decisions happen, we argue that it should give meaningful limitations and guidance to the ethical values embedded in the respective algorithms.

3.3 Risk Distribution in Regular Driving

Outside of specific high-risk scenarios, an AV's behavior during regular driving deserves some scrutiny as well (Himmelreich, 2018; Krügel & Uhl, 2024; Cecchini et al., 2023). Behavioral biases in regular driving can, due to the large number of kilometers driven by the same algorithm across a fleet of vehicles, impact the risk exposure of

various groups of people to different extents, and may carry an even greater potential for dire consequences of ethical misalignment.

One example of such a bias could be the vehicle's chosen target speeds. While speed limits typically present an upper bound for the chosen speed of the vehicle, traffic acts usually demand that the speed is adapted to the current driving situation in a reasonable way (see, e.g., section 3 of the German Road Traffic Regulation). Drivers thus have to use their best judgment and implicitly consider the density of traffic, curvature and available width of the road, partial occlusion through parked vehicles, weather conditions, amount of nearby pedestrians and other factors when setting their speed. Another example is the vehicle's lateral position in its lane. The lateral distance to pedestrians on the sidewalk or cyclists on one side of the vehicle versus the distance to other cars, buses or trucks on the other side isn't typically subject to precise regulation (see however, sections 3, 4, 5, 6 and 7 German Road Traffic Regulation regulating speed, distance to other vehicles, overtaking, passing by and using lanes), but still has an impact on the safety of everyone involved.

Both of these examples thus require some implicit, common sense judgment that is challenging to implement in AVs, and may result in unwanted biases and undue risk exposure for certain road user groups. In deciding on a vehicle's target speed or lateral position in the lane, the AV implicitly or explicitly weighs the utility and safety interests of its passengers against the risk-exposure of pedestrians and other vulnerable road users in the vicinity. The lack of regulation on these behavioral decisions constitutes another omission in the current regulations.

The consideration of regular driving scenarios raises several questions that are currently not clearly addressed in the regulations: In the absence of any immediate risk of a collision, can the manufacturer decide to favor their occupants' safety by giving them larger safety-margins to heavier vehicles than to more vulnerable road users? How much freedom do the manufacturers have when it comes to lateral positioning in the lane, target speeds, and similar decisions that on a large scale impact the overall risk exposure of different road user groups? And more generally, may or must an AV's decision-making logic apply different sets of rules in regular driving versus emergency situations, and where is the boundary between the two?

3.4 Machine Learning-Based Approaches

Beyond their use in perception tasks, deep neural networks and related ML-based approaches are being explored for driving tasks and, crucially, trajectory decisions, both as end-to-end systems (Chen et al., 2024; Teng et al., 2023) and in modular architectures for planning and decision-making (Cheng et al., 2024).

When considering the use of ML-based systems in trajectory planning, one needs to take into account that the system's behavior will be shaped by both the training data and/or the simulation environment, as well as the learning algorithms, objective functions, and potential reward structures. Further, due to the the distributed way in which information is stored and processed in neural networks and the sheer number of parameters (typically in the millions or billions), neural networks are typically not directly interpretable by humans. Consequently, the extraction and evaluation of the

decision-making logic, risk management procedures, and ethical principles from and in ML-based systems may prove difficult or infeasible (for modular and end-to-end systems alike), and demands for transparency and traceability, e.g., after a collision, may be equally difficult to fulfill (Adamson et al., 2019; Luetge, 2017; Birnbacher & Birnbacher, 2016; Teng et al., 2023; De Freitas et al., 2021). On the other hand, it is conceivable for ML-based systems to run with "safety wheels", i.e., handle all regular driving but hand over control to systems with more rigorous safety procedures when certain risk thresholds are surpassed, or to use excessive testing to ensure conformity with behavioral and ethical demands across a wide range of possible scenarios.

Importantly, however, the use of ML-based systems does not diminish the validity of our thesis because a precise definition of behavioral rules or boundaries is required for these systems just the same. In the following, we illustrate this requirement for the two most common ML training paradigms in trajectory planning systems, imitation learning and reinforcement learning, as well as for verification procedures.

In imitation learning, real-world data from human drivers is used, and their trajectory choices are treated as optimal, pushing the behavior of the algorithm towards the dataset average. Since human driving in reality isn't actually always optimal, this targeted dataset average entails a certain propensity for suboptimal behavior such as rule breaking, risky maneuvers, and even causing the occasional accident. To avoid such issues, the data needs to be filtered to only contain behavior deemed correct or acceptable, requiring unambiguous definitions of what constitutes correct or acceptable behavior. If no such definition is available, whoever is in charge of the filtering will inadvertently decide on the algorithm's behavioral policies via their selection of "correct" example cases. The filtering process has further pitfalls: If we assume that all data containing rule violations is filtered out, then those scenarios where rule violations are essentially unavoidable will be filtered out as well. A scenario like *The Stranded Vehicle*, where crossing over a solid line appears as the only reasonable option, would then constitute an "out of distribution" example, i.e., a situation the algorithm has never encountered and wasn't trained to handle, making its behavior unpredictable. Arguably, numerous examples of "correct" behavior both with and without traffic rule violations would be needed for the algorithm to correctly infer and model the underlying behavioral policies – in this case the weighing of utility considerations versus traffic rules. Unfortunately, even large datasets may not contain sufficiently many edge-cases for an algorithm to correctly and precisely infer all underlying behavioral principles from real-world examples.

The issue of insufficient edge-cases can potentially be avoided in reinforcement learning approaches where the algorithm is trained in a simulation environment, receiving precisely defined rewards for desired behavior and punishment for undesired behavior. This reward structure can be thought of as representing an underlying, mathematically precisely defined value system, designed by the manufacturer or developer of the algorithm. The simulation environment allows for any number of edge-cases with numerous variations to be constructed as part of the training procedure, teaching the underlying value system to the algorithm via the reward structure. Over the course of training, the neural network will (ideally) internalize this underlying value system as imprints in its network weights. Clearly, the value system taught to the algorithm during training must be aligned with the set of ethical principles defined by the

regulatory body, and this, once again, requires unambiguous definitions of these ethical principles. Without such definitions, whoever is in charge of defining the reward structure, and with that the underlying value system, will inadvertently decide on the algorithm's behavioral policies (provided that the training is successful).

Of course, a successful training with regards to behavioral principles in edge-cases is not a given. Even if the reward structure and the associated value system are in line with regulatory demands, the use of reinforcement learning by itself doesn't guarantee that this value system is learned correctly and as intended. Compliance with regulatory demands for ethical behavior can thus neither be attested via an analysis of the program code itself, nor by an analysis of the value system it was taught during training. Ensuring that the neural network has indeed learned a behavioral policy in line with the demanded ethical principles may therefore require a large number of simulation-based tests. These could, for instance, determine statistically if personal characteristics of the involved people play a role in the algorithm's decisions, or whether traffic rule violations occur only with sufficient justification in terms of safety or utility. The construction of such test scenarios again requires a precise definition of what constitutes ethically acceptable behavior.

We can generalize from these insights. As established in the introduction, the ethical relevance of any given scenario does not depend on the capacity of the decision-making system to take relevant aspects (implicitly or explicitly) into account. By extension, any system in an AV that is responsible for trajectory decisions – ML-based or otherwise – can be understood as expressing some value system in its behavioral logic. Put another way, the set of trajectory decisions a system produces is inevitably an expression of some internal value system, regardless of how the decisions were reached, the kind of programming and/or training used, or the original intention behind the algorithm (Klenk, 2021; Verbeek, 2006). Thus, even a system producing completely random trajectories would be interpreted as having a value system – in this case one according to which "nothing matters". Similarly, if a system disregards an ethically relevant variable (e.g., the vulnerability of involved road users), this disregard for the variable in question is then part of its inherent value system. Notably, the system's behavioral principles and the ethical value system they express are ultimately defined or at least approved by a human or group of humans. This could be an engineer conceiving of and implementing a deterministic behavioral system, it could be a person collating examples into a training set from which the system learns its behavior, or a committee defining test scenarios for a verification procedure. We thus conclude that any trajectory generating algorithm in an AV inherently represents some specific and logically-numerically precise value system that was designed and/or approved by humans. Making sure that this value system is aligned with the value system envisioned by the regulatory body, or acts within the confines of what the regulatory body deems ethically acceptable, requires well defined behavioral rules or boundaries as part of the legislature.

3.5 Keeping Ethical and Safety-Related Decisions in the Public Domain

The illustrated omissions and imprecision in the existing law leave ample room for interpretation by the manufacturers. In this section we show why this is highly prob-

lematic and argue that it is the responsibility of lawmakers to adequately regulate the behavior of AVs, and in particular trajectory planning.

Article 2 of the European Convention on Human Rights[4] establishes the legal duty of states to protect the lives of those under their jurisdiction and minimize fatal risks. Therefore, the question how an AV makes decisions pertaining to the life and death of their occupants or other road users must be answered by the law and cannot be left up to car manufacturers or private individuals (Kirchmair, 2023c). Since the decision-making logic of AVs is part of their code, it is defined long before any emergency situation arises, and thus fundamentally regulatable. We argue that because it can impact decisions over life and death and is fundamentally regulatable it must be regulated.

This is of particular importance, as manufacturers have a clear incentive to prioritize the safety of their vehicles' occupants over external road users. Bonnefon et al. (2016) found that while participants approved of AVs that sacrifice their passengers to save others, they would not want to buy or drive such vehicles, showing a clear self-preservation bias. Prioritizing the vehicle's occupants over other road users can thus be expected to have a positive impact on sales. As pointed out above, while the current regulations prohibit blatant favoritism in such situations, the vague language may offer loopholes via other permissible parameters that may systematically favor the AV's occupants (e.g., blame for causing the dangerous situation).

Setting aside favoritism towards their customers, manufacturers have a second clear incentive; minimizing their own legal liability, even if this is to the detriment of public safety. In practice, this may lead to very passive behavior of AVs in dangerous situations. Take for example a situation where a pedestrian steps into an AV's lane and swerving around them entails a small risk of minor material damages. Staying in its lane and breaking might endanger the life of the pedestrian but may still carry less risk of legal repercussions for the manufacturer. Under the current law, a severe collision that is somebody else's fault may thus be favorable over a less consequential collision with unclear liability. This can also be detrimental for the AV's occupants: In *The High Stakes Wager* scenario, the driver of the oncoming vehicle is clearly at fault for the dangerous situation. A strategy to minimize legal liability on the side of the manufacturer might therefore favor the passive option of braking and staying in the lane, despite this option bearing a greater risk of physical harm to everyone involved than the alternative of steering the vehicle off the road. Notably, this is still the case if we increase the probability of the other car swerving into the AV's lane for a head-on collision to 50% or even 90% – since the fault is strictly with the other party, the manufacturer's liability may still be minimized by a strictly passive driving strategy. Thus, under the current law, any sensible risk management and collision mitigation might have lower priority than liability considerations. We argue that situations where manufacturers have to decide between the safety of road users and their legal liability should be strictly avoided.

Finally, the compliance with regulations and certifications can, by design, be used by manufacturers as part of a defense against accusations or legal action. However,

[4] Convention for the Protection of Human Rights and Fundamental Freedoms (ETS No. 005), Rome 04/11/1950, entry into force 03/09/1953

when the legal framework is incomplete, imprecise, and ineffective, it can be used as a shield against justified accusations without establishing any tangible safety or ethical standards in return. It is therefore important for the legal framework to effectively regulate its subject matter, which in our opinion, is not currently the case.

3.6 The Need for a Regulatory Framework

We take the safety of all road users and overall fairness in terms of safety between different road user groups to be important objectives for a legal framework for trajectory decisions in road traffic. Therefore, a good legal framework will deem an algorithm with significant impact on the risk-exposure of a large number of people legal only if it makes ethically acceptable decisions, and does not foreseeably expose anyone to undue amounts of risk.

Given the highlighted shortcomings and omissions of the existing set of regulations, we conclude that the current legal framework for trajectory decisions in AVs does not meet these criteria. We therefore argue in favor of a specific legal framework for trajectory decisions in AVs in order to assure safety and fairness for all road users. The regulatory framework should accomplish, among others, the following:

1. Define whether or not different sets of behavioral rules may or must be applied under different circumstances, and define those circumstances (e.g., treating emergency maneuvers separately from regular driving with a defined risk-threshold).
2. Define to which extent decision-making transparency is required (e.g., requirements for traceability of trajectory decisions and human interpretability in case of an accident).
3. Define which factors and/or ethical claims may, must, or must not be taken into account by the algorithm (e.g., personal characteristics, vulnerability, responsibility for the dangerous situation, etc.).
4. Establish a basis for the the balancing of various ethical claims and enable the comparison of potential harm to humans, property damage, traffic rules, considerations of utility and further factors (e.g., by establishing a unified value scale).
5. Provide rules for and limits to the balancing of ethical claims (e.g., if balancing happens via numerical risk-cost-comparisons, the sum of all considered material values could be capped below the value of a severe human injury in order to strike a balance between the prioritization of human well-being and some proportionality in the avoidance of property damages).

In all of this the framework should take into account the incomplete and imprecise assessment of any given situation by the vehicle, the probabilistic nature of any predictions made as part of the decision-making process, and the logical-numerical nature of the algorithms. It should further resolve or avoid internal contradictions, and reduce ambiguities or point to specific standards that define the demands in a more precise way (these standards would then become de facto part of the framework).

We believe that in order to accomplish all of this, it may be the lawmakers' best course of action to formulate the central behavioral demands and limitations in logical-numerical terms, as opposed to the typically abstract and human focused formulations of legal language. Interestingly, an example of numerically precise regulation can

already be found in the EU's type approval regulations: Annex III Part 1 defines the conditions that need to be met before initiating a turning maneuver at an intersection. To this end, whether the AV is allowed to merge into the target lane depends on the fictitious time to collision (TTC) between itself and any approaching privileged (meaning right-of-way) vehicle whose lane may be crossed or joined. The TTC for crossing over a lane with oncoming traffic is, for instance, defined as $TTC_{int} = \frac{v_c}{2 \cdot \beta} + \rho$ with "v_c equal to the speed of the privileged conflicting traffic", "β equal to $3m/s^2$ being the maximum admissible deceleration for the privileged crossing traffic", and "ρ equal to 1.5s being the reaction time of the privileged crossing traffic". While the precise formulation and parameterization of this rule is immaterial to our point, it stands in stark contrast with ethically more relevant rules such as "The ADS shall detect and respond appropriately to objects and events relevant for the [Dynamic Driving Task] within the [Operational Driving Domain]" (followed by a short list of exemplary objects and events) (Annex II, Section 1.2 Commission Implementing Regulation (EU) 2022/1426). This example shows that EU lawmakers are principally willing to (1) give precise numerical limits to the behavior of AVs, and (2) regulate driving behavior directly despite this falling traditionally under the jurisdiction of the member states. Regardless of which jurisdiction regulates AVs behavior we think the laws should have a logical-numerical phrasing similar to the turning and merging behavior described above.

4 Conclusion

In this work we have proposed three ethically challenging traffic scenarios – The Stranded Vehicle (1), The High-Stakes Wager (2), and The Jaywalk Disaster (3) – and conducted an analysis of the current regulations in the EU and Germany that deal with the expected behavior of AVs in these scenarios. This analysis has revealed that the regulations, by and large, do neither guide nor limit the behavior of AVs in a meaningful way. We have highlighted several omissions by the legal text, such as the failure to address the concept of proportionality in the implementation of traffic rules, and the failure to address the distribution of risk exposure in regular driving scenarios. We have further discussed the importance of taking into account the realities of algorithmic decision-making, i.e., imprecise and incomplete information about the status quo, uncertain predictions of future trajectories and outcomes of trajectory options, and the logical-numerical nature of computer code. With regard to the latter, we have highlighted that an interpretation and translation from the abstract legal language into the logical-numerical domain is unavoidable, and that the vague phrasing of the current law leaves excessive room for interpretation. We argued that the required translation must be performed by lawmakers as opposed to car manufacturers, as it is the legal duty of states to ensure that fatal risks are diminished, and the incentives for car manufacturers are not generally aligned with public interest. We therefore urge lawmakers and regulatory bodies to design a comprehensive legal framework that addresses the highlighted omissions and aligns the phrasing of the law with the realities of algorithmic decision-making.

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References

Adamson, G., Havens, J. C., & Chatila, R. (2019). Designing a value-driven future for ethical autonomous and intelligent systems. *Proceedings of the IEEE, 107*(3), 518–525. https://doi.org/10.1109/JPROC.2018.2884923

Awad, E., Dsouza, S., Kim, R., Schulz, J., Henrich, J., Shariff, A., ... Rahwan, I. (2018). The moral machine experiment. *Nature, 563*(7729), 59–64. https://doi.org/10.1038/s41586-018-0637-6

Bigman, Y. E., & Gray, K. (2020). Life and death decisions of autonomous vehicles. *Nature, 579*(7797), E1–E2. https://doi.org/10.1038/s41586-018-0637-6

Birnbacher, D., & Birnbacher, W. (2016). Automatisiertes fahren. ethische fragen an der schnittstelle von technik und gesellschaft. *Information Philosophie, 4*, 8–15.

Bonnefon, J.-F., Černy, D., Danaher, J., Devillier, N., Johansson, V., Kovacikova, T., ... et al. (2020). *Ethics of connected and automated vehicles: Recommendations on road safety, privacy, fairness, explainability and responsibility*. European Commission. https://data.europa.eu/doi/10.2777/035239

Bonnefon, J.-F., Shariff, A., & Rahwan, I. (2016). The social dilemma of autonomous vehicles. *Science, 352*(6293), 1573–1576. https://doi.org/10.1126/science.aaf2654

Cecchini, D., Brantley, S., & Dubljević, V. (2023). Moral judgment in realistic traffic scenarios: Moving beyond the trolley paradigm for ethics of autonomous vehicles. *AI & Society,* 1–12. https://doi.org/10.1007/s00146-023-01813-y

Chen, L., Teng, S., Li, B., Na, X., Li, Y., Li, Z., ... Wang, F.-Y. (2023). Milestones in autonomous driving and intelligent vehicles–part ii: Perception and planning. *IEEE Transactions on Systems, Man, and Cybernetics: Systems, 53*(10), 6401–6415. https://doi.org/10.1109/TSMC.2023.3283021

Chen, L., Wu, P., Chitta, K., Jaeger, B., Geiger, A., & Li, H. (2024). End-to-end autonomous driving: Challenges and frontiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. https://doi.org/10.1109/TPAMI.2024.3435937

Cheng, J., Chen, Y., & Chen, Q. (2024). Pluto: Pushing the limit of imitation learning-based planning for autonomous driving. *arXiv preprint* arXiv:2404.14327, https://doi.org/10.48550/arXiv.2404.14327

De Freitas, J., Censi, A., Walker Smith, B., Di Lillo, L., Anthony, S. E., & Frazzoli, E. (2021). From driverless dilemmas to more practical commonsense tests for automated vehicles. *Proceedings of the National Academy of Sciences, 118*(11), e201020211. https://doi.org/10.1073/pnas.2010202118

Evans, K., de Moura, N., Chauvier, S., Chatila, R., & Dogan, E. (2020). Ethical decision making in autonomous vehicles: The av ethics project. *Science and Engineering Ethics, 26*, 3285–3312. https://doi.org/10.1007/s11948-020-00272-8

Faulhaber, A. K., Dittmer, A., Blind, F., Wächter, M. A., Timm, S., Sütfeld, L. R., ... König, P. (2019). Human decisions in moral dilemmas are largely described by utilitarianism: Virtual car driving study provides guidelines for autonomous driving vehicles. *Science and Engineering Ethics, 25*, 399–418. https://doi.org/10.1007/s11948-018-0020-x

Foot, P. (1967). The problem of abortion and the doctrine of double effect. *Oxford, 5*, 5–15. https://doi.org/10.1093/0199252866.003.0002

Geisslinger, M., Poszler, F., & Lienkamp, M. (2023). An ethical trajectory planning algorithm for autonomous vehicles. *Nature Machine Intelligence, 5*(2), 137–144. https://doi.org/10.1038/s42256-022-00607-z

Goodall, N. J. (2014). Ethical decision making during automated vehicle crashes. *Transportation Research Record, 2424*(1), 58–65. https://doi.org/10.3141/2424-07

Goodall, N. J. (2016). Away from trolley problems and toward risk management. *Applied Artificial Intelligence, 30*(8), 810–821. https://doi.org/10.1080/08839514.2016.1229922

Gruyer, D., Magnier, V., Hamdi, K., Claussmann, L., Orfila, O., & Rakotonirainy, A. (2017). Perception, information processing and modeling: Critical stages for autonomous driving applications. *Annual Reviews in Control, 44*, 323–341. https://doi.org/10.1016/j.arcontrol.2017.09.012

Harris, J. (2020). The immoral machine. *Cambridge Quarterly of Healthcare Ethics, 29*(1), 71–79. https://doi.org/10.1017/S096318011900080X

Henschke, A., & Arora, C. (2024). Pluralism and the design of autonomous vehicles. *Philosophy & Technology, 37*(3), 1–19. https://doi.org/10.1007/s13347-024-00802-3

Himmelreich, J. (2018). Never mind the trolley: The ethics of autonomous vehicles in mundane situations. *Ethical Theory and Moral Practice, 21*(3), 669–684. https://doi.org/10.1007/s10677-018-9896-4

Huang, Y., Du, J., Yang, Z., Zhou, Z., Zhang, L., & Chen, H. (2022). A survey on trajectory-prediction methods for autonomous driving. *IEEE Transactions on Intelligent Vehicles, 7*(3), 652–674. https://doi.org/10.1109/TIV.2022.3167103

International Organization for Standardization (ISO) (2022). *Road vehicles – safety of the intended functionality (sotif, iso 21448:2022)* (Tech. Rep. No. ISO 21448:2022). Geneva, Switzerland: International Organization for Standardization. https://www.iso.org/standard/70939.html. Accessed 22 Dec 2024

Kirchmair, L. (2023a). Autonomous vehicles. *Encyclopedia of the philosophy of law and social philosophy* (pp. 186–192). Springer. https://doi.org/10.1007/978-94-007-6519-1

Kirchmair, L. (2023b). Autonomous vehicles: Crashes. In M. Sellers, & S. Kirste (Eds.), *Encyclopedia of the philosophy of law and social philosophy* (pp. 192–200). Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-007-6519-1_1051

Kirchmair, L. (2023c). How to regulate moral dilemmas involving self-driving cars: The 2021 german act on autonomous driving, the trolley problem, and the search for a role model. *German Law Journal,* 1–25. https://doi.org/10.1017/glj.2023.83

Klenk, M. (2021). How do technological artefacts embody moral values? *Philosophy & Technology, 34*(3), 525–544. https://doi.org/10.1007/s13347-020-00401-y

Krügel, S., & Uhl, M. (2024). The risk ethics of autonomous vehicles: An empirical approach. *Scientific Reports, 14*(1), 960. https://doi.org/10.1038/s41598-024-51313-2

Luetge, C. (2017). The german ethics code for automated and connected driving. *Philosophy & Technology, 30*, 547–558. https://doi.org/10.1007/s13347-017-0284-0

Nyholm, S. (2018). The ethics of crashes with self-driving cars: A roadmap, i. *Philosophy Compass, 13*(7), e12507. https://doi.org/10.1111/phc3.12507

Paulo, N., & Kirchmair, L. (2025). *Moral dilemmas involving self-driving cars: How to regulate them and why your opinion matters*. Taylor & Francis. https://doi.org/10.4324/9781003485216

Paulo, N., & Möck, L. A. (2024). The use and abuse of moral preferences in the ethics of self-driving cars. *Experiments in moral and political philosophy*. Taylor & Francis. https://doi.org/10.4324/9781003301424-18

Paulo, N. (2023). The trolley problem in the ethics of autonomous vehicles. *The Philosophical Quarterly, 73*(4), 1046–1066. https://doi.org/10.1093/pq/pqad051

Poszler, F., Geisslinger, M., & Lütge, C. (2024). Ethical decision-making for self-driving vehicles: A proposed model & list of value-laden terms that warrant (technical) specification. *Science and Engineering Ethics, 30*(5), 47. https://doi.org/10.1007/s11948-024-00513-0

Reed, N., Leiman, T., Palade, P., Martens, M., & Kester, L. (2021). Ethics of automated vehicles: Breaking traffic rules for road safety. *Ethics and Information Technology, 23*(4), 777–789. https://doi.org/10.1007/s10676-021-09614-

Rodríguez-Alcázar, J., Bermejo-Luque, L., & Molina-Pérez, A. (2021). Do automated vehicles face moral dilemmas? a plea for a political approach. *Philosophy & Technology, 34*(4), 811–832. https://doi.org/10.1007/s13347-020-00432-5

Shalev-Shwartz, S., Shammah, S., & Shashua, A. (2017). On a formal model of safe and scalable self-driving cars. *arXiv preprint* arXiv:1708.06374. https://doi.org/10.48550/arXiv.1708.06374

Singer, P. (2023). *Ethics*. Encyclopædia Britannica. https://www.britannica.com/topic/ethics-philosophy. Accessed 5 Aug 2024

Sormoli, M. A., Koufos, K., Dianati, M., & Woodman, R. (2024). A survey on hybrid motion planning methods for automated driving systems. *arXiv preprint* arXiv:2406.05575. https://doi.org/10.48550/arXiv.2406.05575

Sütfeld, L. R., Ehinger, B. V., König, P., & Pipa, G. (2019). How does the method change what we measure? comparing virtual reality and text-based surveys for the assessment of moral decisions in traffic dilemmas. *PLoS ONE, 14*(10), e0223108. https://doi.org/10.1371/journal.pone.0223108

Sütfeld, L. R., Gast, R., König, P., & Pipa, G. (2017). Using virtual reality to assess ethical decisions in road traffic scenarios: applicability of value-of-life-based models and influences of time pressure. *Frontiers in Behavioral Neuroscience, 11*, 122. https://doi.org/10.3389/fnbeh.2017.00122

Teng, S., Hu, X., Deng, P., Li, B., Li, Y., Ai, Y., ... et al. (2023). Motion planning for autonomous driving: The state of the art and future perspectives. *IEEE Transactions on Intelligent Vehicles, 8*(6), 3692–3711. https://doi.org/10.1109/TIV.2023.3274536

Thomson, J. J. (1976). Killing, letting die, and the trolley problem. *The Monist,* 204–217. https://doi.org/10.5840/monist197659224

Verbeek, P.-P. (2006). Materializing morality: Design ethics and technological mediation. *Science, Technology, & Human Values, 31*(3), 361–380. https://doi.org/10.1177/0162243905285847

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